

DBMS Unit 5 — Storage, Indexing & Recovery

Complete Notes

Database Management Systems (DBMS) · Unit 5 · Storage, Indexing & Recovery · 2024

Overview

These notes cover **Storage, Indexing & Recovery**. Design, query and manage relational databases — from ER modelling and SQL to normalization, transactions and recovery. Each section states the core idea, a worked intuition and the points examiners look for.

1. File Organization

Heap, sequential and hashed file organizations determine how records are placed on disk blocks.

Why it matters: File Organization appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of file organization in one line.
- A small example showing file organization in practice.
- Common mistakes students make and how to avoid them.

2. B+ Tree Indexes

B+ trees keep data in sorted leaf nodes linked together, giving logarithmic search and efficient range scans.

Why it matters: B+ Tree Indexes appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of b+ tree indexes in one line.
- A small example showing b+ tree indexes in practice.
- Common mistakes students make and how to avoid them.

3. Hash Indexes

Static and extendible hashing map search keys to buckets for constant-time equality lookups.

Why it matters: Hash Indexes appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of hash indexes in one line.
- A small example showing hash indexes in practice.
- Common mistakes students make and how to avoid them.

4. Log-Based Recovery

Write-ahead logging records undo and redo information so committed work survives crashes.

Why it matters: Log-Based Recovery appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of log-based recovery in one line.
- A small example showing log-based recovery in practice.
- Common mistakes students make and how to avoid them.

5. Checkpoints

Checkpoints bound the amount of log that must be scanned during recovery.

Why it matters: Checkpoints appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of checkpoints in one line.
- A small example showing checkpoints in practice.
- Common mistakes students make and how to avoid them.

6. ARIES

ARIES recovery runs analysis, redo and undo passes using log sequence numbers and compensation log records.

Why it matters: ARIES appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of aries in one line.
- A small example showing aries in practice.
- Common mistakes students make and how to avoid them.

Quick revision

File Organization · B+ Tree Indexes · Hash Indexes · Log-Based Recovery · Checkpoints · ARIES